Colin Vandenhof

colinvdh.com ◇ github.com/ColinVDH

WORK EXPERIENCE

Dash Social, Halifax, Nova Scotia *May 2022 - present*
Senior Research Scientist May 2026 - present
Research Scientist May 2022 - May 2026
Research and development of AI features for a social media marketing platform, including a conversational agent for social analytics, brand-aware image and video generation, creator demographic prediction, and tools for surfacing trends and topics in social listening. Built evaluation pipelines to benchmark models for each.

Spiri Robotics, Halifax, Nova Scotia *February 2021 - December 2021*
Software Engineer
Development of systems for data analysis and machine learning. Implemented Django REST API for managing datasets, training scripts, and models. Integrated Slurm for job scheduling. Created dashboards for visualizing aggregated flight data. Server provisioning with Ansible, and container orchestration with Docker Swarm.

University of Waterloo, Waterloo, Ontario *January 2018 - December 2019*
Teaching Assistant

- Fall 2019, Winter 2019, Spring 2018 - CS 486/686: Introduction to Artificial Intelligence
- Winter 2019 - CS 480/680: Introduction to Machine Learning
- Winter 2018 - CS 136: Elementary Algorithm Design and Data Abstraction

Responsible for holding office hours, answering questions on course forums, proctoring, and marking.

Element AI, Toronto, Ontario *May 2019 - August 2019*
Applied Research Intern
Conducted research into uncertainty quantification, and the use of utility models for decision-making in production deep learning systems. Model development and experimentation in Python using PyTorch. Creation of interactive data visualization tools in JavaScript using D3.js.

Blockcrushr Labs, Halifax, Nova Scotia *September 2017 - December 2017*
Software Developer
Development of universal web application for charitable giving backed by the Ethereum blockchain. Front-end web development in React. Implemented user authentication using JWT. Integration with Node.js back-end.

Atlantic Lottery Corporation, Halifax, Nova Scotia *June 2017 - August 2017*
Summer Student - Software Development
Development of Amazon Alexa application with Node.js and deployment with AWS. Development of web scraper in Javascript and deployment of lottery results RESTful API with AWS. Integration of API with Ruby on Rails web application. Front-end web development in React.

Dalhousie University, Halifax, Nova Scotia *May 2017 - August 2017*
Teaching Assistant

- Spring 2017 - CSCI 3120: Operating Systems

Responsible for editing assignments, leading in-class activities, proctoring, and marking.

EDUCATION

University of Waterloo *January 2018 - May 2020*
Master of Mathematics (Computer Science) GPA: 4.00 / 4.00

- Canada Graduate Scholarship - Doctoral ($105000, 2020, declined)
- Alexander Graham Bell Canada Graduate Scholarship ($17500, 2019-2020)
- President's Graduate Scholarship ($5000, 2019-2020)
- David R. Cheriton Graduate Scholarship ($20000, 2018-2019)
- Mathematics Domestic Graduate Student Award ($4000, 2018-2019)

Dalhousie University *September 2016 - April 2017*
Special Undergraduate of Science (Computer Science) GPA: 4.00 / 4.00

- MacKenzie Scholarship ($2000, declined)

McMaster University *September 2012 - April 2016*
Bachelor of Health Sciences (Honours) GPA: 3.94 / 4.00

- Provost's Honour Roll Medal (2016)
- President's Award ($2500, 2012)

PUBLICATIONS

A Hybrid Approach to Identifying Unknown Unknowns of Predictive Models. Vandenhof, C. In Proceedings of the 7th AAAI Conference on Human Computation and Crowdsourcing.

Contradict the Machine: a Hybrid Approach to Identifying Unknown Unknowns. Vandenhof, C. & Law, E. In Proceedings of the 18th International Conference on Autonomous Agents and Multiagent Systems.

TECHNICAL SKILLS

Programming Languages

Proficient:	Python, JavaScript, Bash
Previous experience:	C/C++, Java, R, MATLAB

Software and Frameworks

Version control:	Git, SVN
Web technologies:	React, HTML/CSS, Flask, Node.js, PostgreSQL